RESEARCH REPORT · VITKI

Methods report, partially tested

Statistical carver suggestions from the runic texts: 82 percent right among the confident suggestions in a locked blind test, against 44 percent for the nearest carver

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Report version 4 · Published 31 October 2026 · Analysis in Vitki 3.0.0 · Licence CC BY 4.0

Last changed 8 November 2026

Archived on Zenodo: **DOI 10.5281/zenodo.24054744**

Timeline of the study: The model was built 5 October 2026 · The plan of the blind test was locked 21 October 2026 · The blind test was run 21 October 2026 · The test was rerun on a later version of the material 31 October 2026

ABSTRACT

What has not been tested. The model behind the suggestions was changed on 4 November 2026, after report version 1, and the blind test has not been rerun on the changed model, which gives 106 confident and 313 possible suggestions instead of 86 and 292; the figures in this report apply to the model as it was 21–31 October 2026 (section 8). The blind test was locked in advance and each stone's carver was hidden, but the weights and limits of the suggestion rule were set on the same material, and a test on stones whose carver the rule has not seen (attributed in Runor 2020 but not in Rundata) is too small: 8 stones.

Background and aim. Rune carvers are attributed by runologists from ornament, rune forms, spelling and place, and since the 1990s also statistically from the shape of the grooves in laser scans. The question here is how often a statistical rule that uses only what the runic-text database contains suggests the right carver when the carver is hidden.

Material. 769 Swedish Viking-Age runestones with a secure carver in the Scandinavian Runic-text Database (Rundata); 28 carvers can be suggested.

Method. A blind test locked before it was run: for each stone the carver is hidden, all carver profiles are recomputed without it, and the suggestion is compared pairwise with a rule that suggests the carver of the nearest stone with a known carver (McNemar's exact test). The suggestions rested on a cross-validated random-forest model of language, content, style and place, and on a comparison of spelling.

Results. The rule gave a confident suggestion for 302 stones and was right on 248 (82 percent, 95 % interval 77–86); the nearest carver was right on 133 (44 percent), $p = 3.1 \cdot 10^{-23}$. On signed stones, where the

carver wrote his own name: 72 of 100 against 42. The advantage is clear in Uppland but not in Södermanland. The same test rerun on the material of 31 October 2026 gave the same result stone by stone.
Conclusion. In the test the suggestions were right about twice as often as a simple guess by place and received a known error rate, so they can be put forward as testable hypotheses. They show above all how well the rule reproduces the runologists' own judgements, and they do not replace a runological examination.

Keywords: *runestones, Viking Age, rune carvers, attribution, random forest, blind test, preregistration, orthography*

HIGHLIGHTS

- A rule that reads only what the database contains can suggest who carved a stone, and when it is confident it is right about four times in five.
- It beats a guess by the nearest stone by a wide margin, also on the stones where the carver wrote his own name.
- The suggestions are hypotheses with a known error rate, best in Uppland, and they do not replace the runologist's examination of the stone.
- The error rate was measured on the model as it was in October 2026; the changed model of 4 November has not been blind-tested.

1. Introduction

Runologists attribute unsigned runestones to carvers from ornament, rune forms, spelling, formulas and place. Claiborne Thompson (1975) showed how spelling tells the Upplandic carvers apart. Jan Axelson (1993) listed the signed and attributed inscriptions of central Sweden, and most attributions in Rundata for Uppland, Södermanland, Västmanland and Gästrikland rest on his list. Per Stille (1999) attributed the stones of Fjädrundaland, and Magnus Källström (2007) has studied the carvers and their writing milieus across Scandinavia.

Statistical attribution is not new. In a method study with a modern carver, Laila Kitzler (1998) showed that the shape of the grooves in laser scans tells carvers apart with at least 70 percent accuracy, using cluster and discriminant analysis. She also tested Frands Herschend's thesis that the signatures ubiR and ybiR/ybir belong to two different carvers named Öpir (Herschend 1998), and found support for it. She has since applied the method to Viking-Age stones, above all in Uppland (Kitzler Åhfeldt 2002), and on Bornholm (Kitzler Åhfeldt 2020). The method requires a scan of every stone.

We have found no method that suggests carvers for all runestones from what the runic-text database contains and that has been tested in a blind test locked in advance, against a simple comparison rule on the same stones. The aim here is to report such a test.

2. Pre-registration: the plan of the blind test

The plan was written and locked on 21 October 2026, before the blind test was computed. It is time-stamped in the project's version control and has the SHA-256 checksum `5ed0b2f3b147aad4…` (see Data and code). The question in the plan reads: are the suggestions with the status *new suggestion* right clearly more often than a simple rule that suggests the carver of the nearest stone with a secure carver, on the same stones? *Yes* if McNemar's test gives p below 0.05. A no would be reported just as clearly. The plan forbids changes to weights, limits and rules after the results have been seen.

Two things were known before the plan was locked: how many stones without a carver received each status in the project's overview, and which stones have a carver in the Swedish National Heritage Board's register Runor (2020) but not in Rundata. The suggestion rule and the model were locked in the version tested. But they had been developed on the same material: the model's limit for "perhaps someone else" was chosen with stones by carvers left out of the model, and three details of the rule were adjusted on 11 October 2026 after examination of individual stones (U 92, U 735 and U 1026). The answer was thus hidden for every stone in the test, but not when the rule was designed. That is why the report is *partially tested*.

Sections 5.1 and 5.4 are the plan's tests. Sections 5.2 and 5.3 are additions beyond the plan.

3. Material

- **Source.** The Scandinavian Runic-text Database (Rundata) 3.1, Department of Scandinavian Languages, Uppsala University: transliteration, normalisation, province, parish, place, carver, style group and rock type for each inscription.
- **The stones in the test.** All 769 Swedish Viking-Age runestones with a secure carver in Rundata, signed or attributed without a question mark. 636 have a carver who can be suggested. 133 have a carver who cannot be suggested; they cannot be right but are included, so that it shows how often a wrong carver is suggested.
- **The carvers who can be suggested.** 28: the 18 carvers the model is trained on (at least 8 secure stones) and the carvers who have a spelling profile (at least five secure inscriptions).
- **Runestones only.** Other objects with runes are not included.

4. Method

4.1 The model. A random forest (Breiman 2001) with 500 trees, trained on 497 stones by 18 carvers. The variables were location and province, style group, cross and short-twig runes, eleven language traits, five formulas, the hundred most common rune pairs and nine content categories. The model was evaluated by fivefold cross-validation repeated three times, so that each stone's probabilities came from models that did not see it, and also with whole parishes and whole hundreds left out. If the first choice's probability was below 0.35, the model said *perhaps someone else*. The model gave

probabilities only for stones within 25 km of a training stone. This is the model as it was when the test was run; it has been changed since (section 8).

4.2 The spelling. Each inscription is compared with the carvers' spelling profiles: parts of words of two and three runes, word dividers and bind-runes, without proper names, so that a signature does not give the answer away. The stone itself is not counted in the profile.

4.3 The suggestion rule. The strength of a suggestion is computed from the model's probability and the spelling's first choice and lowered by four checks: whether the stone stands within the carver's area, whether the style group is among the carver's usual ones, whether the language traits agree and whether the rock type agrees. A *new suggestion*, here called a confident suggestion, is given at a strength of at least 0.5 and a *possible suggestion* at at least 0.3.

4.4 The blind test. For each stone in the test the carver is made unknown in a copy of the material, and the carvers' areas, style groups, rock types and language profiles are recomputed without the stone. The suggestion is compared with the rule *nearest carver*: the carver of the nearest other stone with a secure carver, all carvers if several stones stand at the same distance. The measures are the proportion correct with Wilson's 95 % interval and the pairwise comparison with McNemar's exact test. An independent check recomputed 60 randomly chosen stones in a different way and got the same outcome for all of them.

4.5 Deviations and rerun. Three deviations from the plan. A coding error cut the list of the model's suggestions to the first 400 stones; it was corrected after the first run. That first run already gave 247 right of 300 confident suggestions, 82 percent, against 41 percent for the nearest carver. The nearest-carver rule was then made generous at equal distances, which makes the comparison harder for the suggestion rule. The breakdown by province and carver in 5.2 is an addition. On 31 October 2026 the test was rerun unchanged on the material of that day; the model's probabilities were identical and the outcome was the same for every stone.

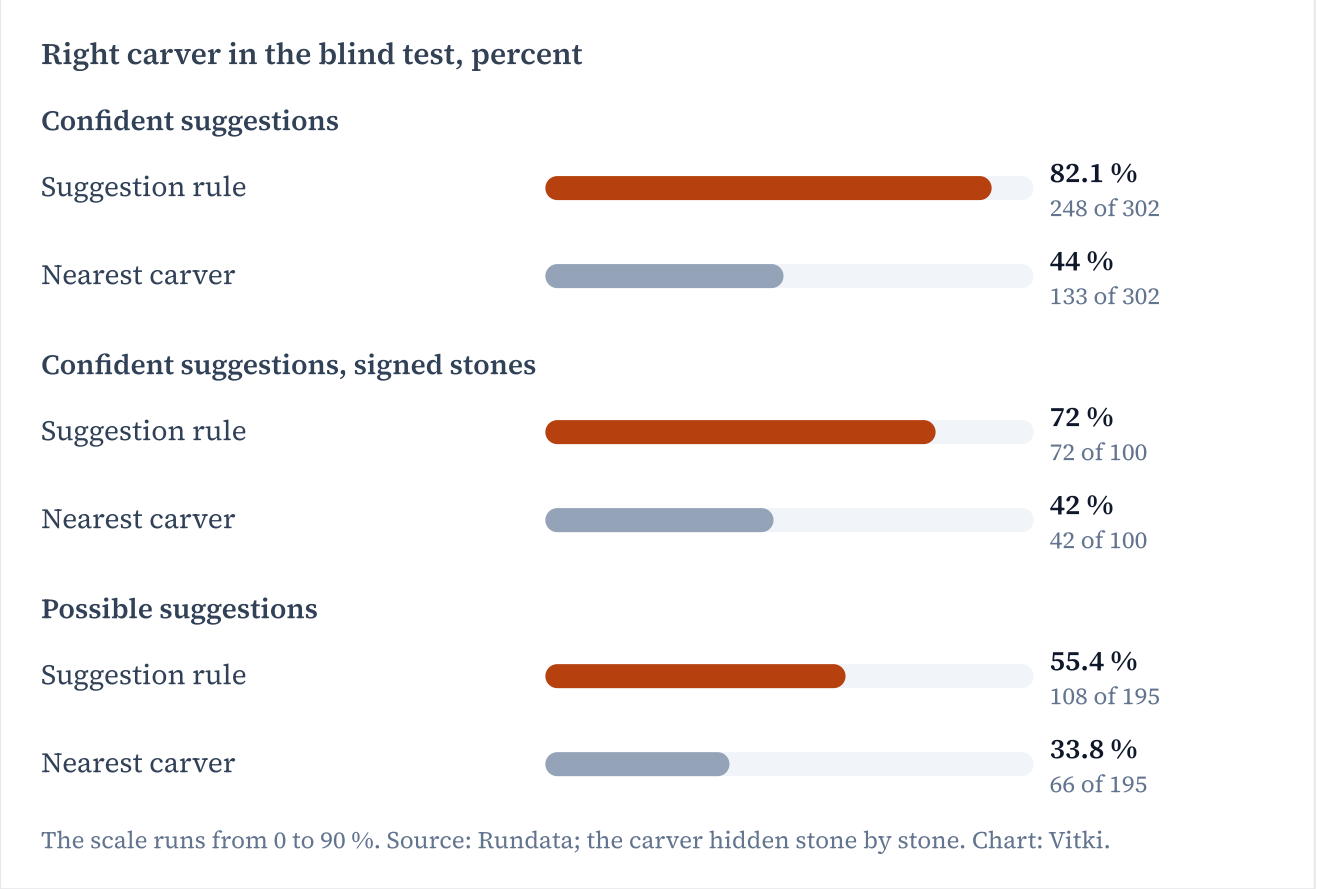
5. Results

5.1 The blind test. Of the 769 stones, 302 received a confident suggestion, 195 a possible one and 272 none.

- **Confident suggestions:** right on 248 of 302 (82 percent, 77–86). Nearest carver: 133 (44 percent). On 132 stones only the suggestion rule was right, on 17 only the simple rule: $p = 3.1 \cdot 10^{-23}$. The answer to the plan's question is *yes*.
- **Signed stones**, where the answer key is the carver's own name: 72 of 100 (72 percent) against 42, $p = 9.2 \cdot 10^{-6}$.
- **When the model gives a name:** 213 of 254 (84 percent). When it says *perhaps someone else*: 35 of 48 (73 percent).
- **Possible suggestions:** 108 of 195 (55 percent) against 66 (34 percent).
- **By strength:** 0.5–0.6 right on 50 of 72; 0.6–0.7 on 56 of 65; at least 0.7 on 142 of 165.

- **Carvers outside the 28:** 23 of the 133 stones still received a confident suggestion, which is necessarily wrong; they are included in the figures above.

In plain words: when the rule gives a confident suggestion it is right about four times in five, and a guess by the nearest stone is right just over two times in five.



5.2 By province and by carver (beyond the plan). Uppland: 227 of 270 (84 percent) against 43 percent, $p = 1.2 \cdot 10^{-23}$. Södermanland: 17 of 28 (61 percent) against 13 (46 percent), $p = 0.39$, so no clear difference. By suggested carver with more than three confident suggestions: Öpir 1: 68 of 104; Åsmund: 37 of 39; Balle: 34 of 37; Fot 2: 30 of 35; Gunnar: 21 of 22; Traen: 11 of 15; Erik: 14 of 14; Livsten: 10 of 10; Arbjörn: 7 of 7; Visäte: 6 of 6; Torbjörn 2: 4 of 4. Öpir has the lowest proportion correct. All four errors for Traen (Sö 46, Sö 159, Sö 160, Sö 296) concern stones signed by other carvers in Södermanland.

5.3 The model on its own (the cross-validation). The model alone put the right carver first on 72.8 percent of the training stones and among the first three on 86.7 percent; always guessing the most common carver gives 19 percent. With whole parishes left out: 68.8 percent; with whole hundreds: 58.6 percent.

- **What carried the model.** Language and content only: 61.4 percent; geography only: 43.9; style only: 33.0. As a comparison level, not as part of the model: rock type alone gives 18.9. With whole hundreds left out, language held up better than geography, 50.1 against 30.2 percent. Language and content accounted for 72 percent of the variables' combined importance.

- **The calibration.** The model was cautious: when it gave about 25 percent to its first choice it was right in 51 percent of cases, at 45 percent in 81 and at 65 percent in 98. The probabilities thus underestimated the accuracy, and they should be read as a ranking rather than as odds.
- **Perhaps someone else.** The limit 0.35 flagged 80 percent of the stones when their carver had been left out of the model, but also 41 percent of the known carvers' stones; among the stones not flagged, the first choice was right in 86.9 percent.
- **The confusions.** Most often Åsmund (10 stones), Gunnar (8) and Visäte (8) were taken for Öpir, and Tidkume for Balle (10).

5.4 External test (the plan's secondary). 19 runestones have a named carver in Runor (2020) but none in Rundata; 8 of them have a carver who can be suggested. None received a confident suggestion, 3 received a possible one, and 1 of those was right. That is too few stones to say anything, as the plan expected.

6. Discussion

What the method can do. When the suggestion rule gave a confident suggestion, it was in the test right about four times in five, twice as often as a guess by the nearest stone. That holds also for signed stones, where the answer key is not a judgement but the carver's own name. The 86 confident and 292 possible suggestions the rule gave on 21 October 2026 for the 1,550 stones without a carver therefore carry a known error rate, and they can be put forward as hypotheses to be tested against the stones themselves. The suggestions the changed model gives have not been tested in the same way (section 8).

Against earlier research. Kitzler Åhfeldt's method and this one measure different things. Hers rests on how the carver drove the chisel and requires a scan; this one rests on what the carver wrote and where the stone stands, and works for every stone in the database. That other carvers' stones are most often taken for Öpir probably reflects his being the largest class in the model (95 of 497 training stones). Rundata moreover assigns all his signatures, both ubiR and ybiR/ybir, to one and the same carver, Öpir 1 (Rundata's Öpir 2 is another carver, in Södermanland), although Herschend (1998) argued that they are two carvers and Kitzler (1998) found support for this in the grooves. If that is right, the model's largest class mixes two hands. The two methods are independent of each other, and where both can be used they test each other.

Weaknesses, in order of weight.

1. **The answer key is largely judgements of the same kind.** Of the 769 stones, 523 are attributed, not signed, and the attributions rest on style, spelling and place, in central Sweden largely via Axelson (1993). The test therefore partly measures how well the rule reproduces the runologists' judgements. A test after report version 3 confirms this: against the runologists' own judgements in Runor 2020 of which stones were made by the same hand, also when the hand has no name, the model did not give the same carver within such a group more often than neighbouring stones at the same distance get (50 against 44 per cent of the stone pairs in 14 groups, better in 5 and worse in 7; plan locked 8 November 2026). The proportion correct thus measures agreement with Rundata's attributions, not the ability to recognise a hand. The text features themselves do

carry a hand beyond geography (against the same judgements: 16 of 18 groups more alike than the neighbours), which suggests that the fault is that the rule can only choose among named carvers. The signed stones are the independent test, and there the proportion correct is lower: 72 percent.

2. **The rule was set on the same material.** Weights and limits were locked before the test but chosen with the material in view (section 2). That probably gives a somewhat too high accuracy. A test on stones whose carver the rule has not seen would remove this, but such a test (attributed in Runor 2020 but not in Rundata) is too small: 8 stones.
3. **Only 28 carvers.** A stone by another carver can only receive a wrong suggestion. 23 such stones received a confident suggestion. Among the stones without a carver there are surely many by carvers the rule does not know.
4. **Södermanland.** The advantage is not clear there (17 of 28 against 13), and almost all confident suggestions concern Uppland.
5. **Öpir.** The largest carver in the model attracts stones by other carvers, and the proportion correct is lowest for him (68 of 104).

Where the method should not be used. Outside Uppland, for stones more than 25 km from a stone with a known carver, for carvers with few signed stones, to decide between neighbouring stones in one and the same carving milieu, and as proof of an attribution. A suggestion is a hypothesis.

What would overturn the method. If the proportion correct fell to the nearest-carver level on stones attributed after the rule was locked, or if groove analyses of scanned stones contradicted the confident suggestions.

7. Conclusion

A statistical suggestion rule that uses only what the runic-text database contains suggested, in a locked blind test, the right carver for 82 percent of the stones where it gave a confident suggestion, against 44 percent for the nearest carver, and for 72 percent of the signed stones. In the test it worked as a way of sorting out testable attribution hypotheses in Uppland. The next step is to rerun the blind test on the changed model, a test on stones attributed after the rule was locked, and a comparison with groove analyses of the scanned stones.

8. The model after report version 1

After report version 1 the model changed on 4 November 2026: cross forms from Runor (2020) were added as features, calibrated probabilities were added in separate fields, and the *perhaps someone else* limit became 0.335, chosen on probabilities computed with whole parishes left out. The suggestion rule still computes its strength from the forest's uncalibrated probability, with the same limits (0.5 and 0.3), but that probability now comes from a new forest, and the blind test has not been rerun on it.

- **The changed model** put the right carver first, in the run of 4 November, for 74.0 percent of the training stones and among the first three for 89.1 percent; with whole parishes left out 69.2 percent and with whole hundreds 60.6. Style alone gave 53.9 percent instead of 33.0, because the cross forms are counted there.
- **Against the signed stones.** The same day, before the change, the model of this report was tested under a locked plan against the 117 signed stones among the training stones, with whole parishes left out and on a later version of the material. The first choice was right for 71.2 percent (63.9–78.6), but averaged per carver only for 43 percent, and for six carvers with one to four signed stones no first choice was right. With the cross forms added the proportion was the same, 71.2 percent.
- **The suggestions.** With the changed model the rule gives 106 confident and 313 possible suggestions for the 1,550 stones without a carver, against 86 and 292 on 21 October 2026. These suggestions have no blind-tested error rate.

The figures in this report apply to the model as it was 21–31 October 2026.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se), with the changes made after that version up to 31 October 2026 (commit `5095161` in the GitHub repository; the code and results of the blind test last changed in commit `06e5c47` the same day). The model was changed on 4 November 2026 in commits `fe616f4` and `7963a9d`; the files for section 8 and the report's builder for the versions of 2026-11-05 are in commit `6eae0af` onward. All code, the analysis plan and the result files are openly available on GitHub under GPL-3.0.

- The plan of the blind test: `utdata/forslag/analysplan_forslag.md`, with its checksum in `analysis_plan_forslag.sha256`.
- The blind test: `utdata/forslag/blindprov.py` (results `blindprov.json`; the rerun on the material of 31 October 2026, version `e83202e01a5e6d27`, in `blindprov_e83202.json`), the independent check `kontroll.py`, the deviations in `resultat.md`. The suggestions of 21 October 2026: `nya_forslag.json`.
- The model: `r/attribution.R` (results `attribution.json` under the material's version `e83202e01a5e6d27`). The spelling: `src/orthography.py`. The suggestion rule: `src/stone_overview.py`. Method description: `METHODS.md`, sections 1, 7 and 14c.
- Section 8: the changed model and its suggestions in `utdata/attribuering_provad/modell_2026_11_04.json` (script `modell_2026_11_04.py`, the material's version `3691a2c441dd03f4`); the test against the signed stones in `utdata/attribuering_provad/resultat.json` (plan `analysplaner/2026-11-04-attribuering-provad.md`, code `r/attribution_audit.R`); the cross forms in `utdata/attribuering_v2/resultat.json` and `utdata/attribuering_v3/resultat.json`.
- The report is built by `utdata/blogg/build/build_rapport_ristarmodellen.py`, which reads every number from the result files and stops if anything disagrees. The template: `utdata/rapporter/MALL.md`, version 4.
- The current model's suggestions, carver by carver (in Swedish): vitkirunor.se/ristarna (not the model of this report, see section 8). Popular account: Five runestones in search of their carver.

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Change log

Version 2, 2026-11-05. Corrected after a review before the permanent archiving in Zenodo. The blind test, its figures and the conclusion are unchanged. The changes and the reasons for them:

- **The model has changed.** Version 1 described the model as if it were still current. The model was changed on 4 November 2026; this is now stated first in the abstract and in a new section 8 with the changed model's accuracy, the test against the signed stones and the rule's new number of suggestions (106 confident and 313 possible). Sections 4.1 and 5.3 are in the past tense, and the essentials box has a line about it.
- **The suggestions are dated.** The 86 confident and 292 possible suggestions are those the rule gave on 21 October 2026.
- **The external test.** The 8 stones were described as attributed only after the rule was locked. They are attributed in Runor 2020 but not in Rundata and were known before the lock (abstract and weakness 2).
- **Öpir.** Version 1 cited only Kitzler for the thesis that Öpir is at least two carvers and linked it to the rule hitting Öpir least often. The thesis is Herschend's (1998); Kitzler (1998) tested it and

found support for it. The confusions are more simply explained by Öpir being the largest class in the model. The introduction and the discussion now say so, and that Rundata assigns both ubiR and ybiR/ybir to the same carver. Herschend (1998) has been added to the references.

- **Kitzler.** Kitzler Åhfeldt (2002) concerns above all Uppland, not Uppland and Östergötland. Kitzler (1998) is cited with the journal's name when the article was printed, *Laborativ arkeologi* 10–11.
- **Form.** The template version is 4, not 2. The app's tabs and a relative link have been replaced with full web addresses. "Today's material" has been replaced with the material's date and version. The code state is given as a commit. The chart has its credit, and rock type is described as a comparison level, not part of the model.
- **Wording.** "Testable attribution hypotheses" instead of "testable attributions"; "the model gave probabilities only for stones within 25 km" instead of "suggestions are given only ..."; "the proportion correct" instead of "the share right"; "for carvers with few signed stones" has been added under Where the method should not be used.

Version 3, 2026-11-05. Corrected after a second review before archiving in Zenodo. The blind test, its figures and the conclusion are unchanged. The changes and the reasons for them:

- **Kitzler Åhfeldt (2002).** The series is *Theses and Papers in Scientific Archaeology* B:9, as she cites it herself (Kitzler Åhfeldt 2020), not *Theses and Papers in Archaeology*.
- **Herschend (1998).** The reference has been checked against the article (Fornvännen 93, pp. 97–111): he concludes that ubiR is one carver and ybiR/ybir another. Kitzler (1998) cites it as Herschend 1997. The text is unchanged.
- **Code state.** The files for section 8 (modell_2026_11_04.json and .py) and the report's builder are in commit 6eae0af, which Data and code now names.
- **What version 2 did not say.** The abstract's method and conclusion, 4.5, the discussion and the conclusion were also put in the past tense ("In the test the suggestions were ...", "In the test it worked ..."), not only 4.1 and 5.3.

Version 4, 2026-11-08. A new test after the archiving of version 3. The blind test, its figures and the conclusion are unchanged. The changes and the reasons for them:

- **Same hand (weakness 1).** The model was tested against a key it has never seen, the runologists' "same as made X" in Runor 2020, with neighbouring stones at the same distance as the comparison (analysplaner/2026-11-08-samma-hand-runor.md, the result in commit 285b204). The locked plan decided that the outcome should stand among the methods report's limitations; it now stands in weakness 1.
- **The app.** In Vitki the rule's confident suggestions have since 8 November been called *suggestions among known carvers* and the possible ones *weak suggestions*, and no suggestion is called strong without evidence independent of spelling, place and style. The report's terms are unchanged.

Competing interests: none.

Funding: none.

Cite as

Kvant, V. (2026). Statistical carver suggestions from the runic texts: 82 percent right among the confident suggestions in a locked blind test, against 44 percent for the nearest carver. Research report (methods report), report version 4; analysis in Vitki 3.0.0. https://vitkirunor.se/en/forskningsrapporter/ristarforslag-blindprov https://doi.org/10.5281/zenodo.24054744